MTE 203 – Advanced Calculus
Week 3 Homework Solutions

Note:

This Homework covers the material of Lectures 6, 7, 8 and 9 and it is therefore inherently long. You are encouraged to pace yourselves while working through the problems rather than attempting to complete everything in one sitting. Many of the questions involve graphing, which can be time-consuming and occasionally quite intricate, so please plan accordingly and start early.

Parametric and Vector Representations of Curves (S.11.10)

Problem 1 [11.10; Prob. 7]

Find the parametric and vector representation for the curve,

$$z = x + y$$

$$y = x^2$$

directed so that x increases along the curve. Draw the curve indicating its direction.

Solution:

If we set $x = t$, x will be increasing along the curve C, because the parameter t is always increasing in the direction of motion along the curve.

Replacing in the equations of the given surfaces we obtain,

$$y = t^2$$

$$z = t + t^2$$

The parametric equations are,

$$x = t$$

$$y = t^2$$

$$z = t + t^2$$

The vector representation is,

$$\vec{r}(t) = t\hat{\imath} + t^2\hat{\jmath} + (t + t^2)\hat{k}$$

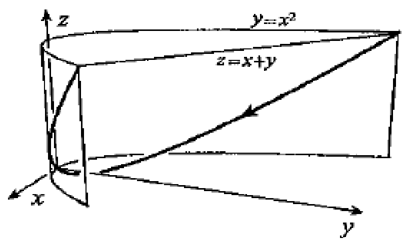
To draw the direction of the curve, we need to plot three points in the curve of intersection for three different values of t,

t =-1, $P_1(-1,1,2)$

$t = 0,\ P_2(0,0,0)$

$t = 1,\ P_3(1,1,2)$

Drawing the curve indicating its direction,



Problem 2 [11.10; Prob. 9]

Find the parametric and vector representation for the curve,

$$x = \sqrt{z}$$

$$z = y^2$$

directed away from the origin in the first octant. Draw the curve indicating its direction.

Solution:

If we set $y = t$, replacing in the equations of the given surfaces we obtain,

$$z = t^2$$

$$x = |t|$$

Where we are taking the absolute value of t because the curve is in the first octant. The parametric equations are,

$$x = |t|$$

$$y = t$$

$$z = t^2$$

The vector representation is,

$$\vec{r}(t) = |t|\hat{\imath} + t\hat{\jmath} + t^2\hat{k}$$

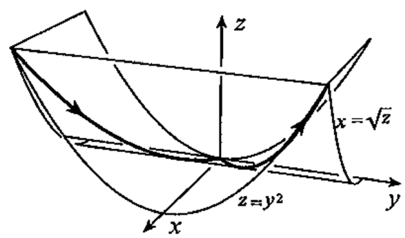
To verify the direction of the curve, which needs to be directed away from the origin, we need to plot three points in the curve of intersection or three different values of t,

t =-1, $P_1(1,1,1)$

$t = 0$, $P_2(0,0,0)$

$t = 1$, $P_3(1,-1,2)$

Drawing the curve indicating its direction,



Differentiation and Integration of Vectors (S.11.9)

Problem 1 [11.9, Prob. 7, 19]

If $f(t) = t^2 + 3$, $g(t) = 2t^3 + 3t$, $\vec{\boldsymbol{u}}(t) = t\hat{\boldsymbol{\imath}} - t^2\hat{\boldsymbol{\jmath}} + 2t\hat{\boldsymbol{k}}$, and $\vec{\boldsymbol{v}}(t) = \hat{\boldsymbol{\imath}} - 2t\hat{\boldsymbol{\jmath}} + 3t^2\hat{\boldsymbol{k}}$, find the scalar or the components of the vector in the following exercises:

7: $\frac{d}{dt}[f(t)\vec{\boldsymbol{v}}(t)]$

19: $\int[f(t)\vec{\boldsymbol{u}} \cdot \vec{\boldsymbol{v}}]dt$

Solution:

7:

$f(t)$ is a scalar function and $\vec{\boldsymbol{v}}(t)$ is a vector function. The derivative follows Theorem 11.9: Trim Chapter 11, Page 758.

$$\begin{aligned}\frac{d}{dt}[f(t)\mathbf{v}(t)] &= f'(t)\mathbf{v}(t) + f(t)\mathbf{v}'(t) = 2t(\hat{\mathbf{i}} - 2t\hat{\mathbf{j}} + 3t^2\hat{\mathbf{k}}) + (t^2+3)(-2\hat{\mathbf{j}} + 6t\hat{\mathbf{k}}) \\ &= 2t\hat{\mathbf{i}} - 6(t^2+1)\hat{\mathbf{j}} + 6t(2t^2+3)\hat{\mathbf{k}}\end{aligned}$$

19:

Evaluate the integrand, noting that $\vec{\boldsymbol{u}} \cdot \vec{\boldsymbol{v}}$ will result in a scalar function.

$$\begin{aligned} f(t)\vec{\boldsymbol{u}} \cdot \vec{\boldsymbol{v}} &= (t^2 + 3)\left(\left(t\hat{\boldsymbol{\imath}} - t^2\hat{\boldsymbol{\jmath}} + 2t\hat{\boldsymbol{k}}\right) \cdot \left(\hat{\boldsymbol{\imath}} - 2t\hat{\boldsymbol{\jmath}} + 3t^2\hat{\boldsymbol{k}}\right)\right) \\ &= (t^2 + 3)(t + 2t^3 + 6t^3) \\ &= 8t^5 + 25t^3 + 3t \end{aligned}$$

The integral can be evaluated as normal.

$$\int (8t^5 + 25t^3 + 3t)dt = \frac{4t^6}{3} + \frac{25t^4}{4} + \frac{3t^2}{2} + C$$

$$\int [f(t)\mathbf{u} \cdot \mathbf{v}]\, dt = \int (t^2 + 3)(t + 2t^3 + 6t^3)\, dt = \int (8t^5 + 25t^3 + 3t)\, dt = \frac{4t^6}{3} + \frac{25t^4}{4} + \frac{3t^2}{2} + C$$

Problem 2 [11.9, Prob. 25]:

Prove that if a differentiable function $\vec{\mathbf{v}}(t)$ has constant length, then at any point at which $\frac{d\vec{\mathbf{v}}}{dt} \neq 0$, the vector $\frac{d\vec{\mathbf{v}}}{dt}$ is perpendicular to $\vec{\mathbf{v}}$**.**

Solution:

We are given $\vec{\mathbf{v}}(t)$ has constant length, which implies $|\vec{\mathbf{v}}(t)| = c$, where c is a constant.

Squaring both sides gives the following equation.

$$\begin{aligned} |\vec{\mathbf{v}}|^2 &= c^2 \\ \vec{\mathbf{v}} \cdot \vec{\mathbf{v}} &= c^2 \end{aligned}$$

Differentiate both sides.

$$\begin{aligned} &\frac{d\vec{\mathbf{v}}}{dt} \cdot \vec{\mathbf{v}} + \vec{\mathbf{v}} \cdot \frac{d\vec{\mathbf{v}}}{dt} = 0 \\ &2\left(\vec{\mathbf{v}} \cdot \frac{d\vec{\mathbf{v}}}{dt}\right) = 0 \\ &\left(\vec{\mathbf{v}} \cdot \frac{d\vec{\mathbf{v}}}{dt}\right) = 0 \end{aligned}$$

Since the dot product of $\vec{\mathbf{v}}$ and $\frac{d\vec{\mathbf{v}}}{dt}$ is 0, the vectors are perpendicular at any point where $\frac{d\vec{\mathbf{v}}}{dt} \neq 0$.

If $\mathbf{v}$ has constant length, then $\mathbf{v} \cdot \mathbf{v} = |\mathbf{v}|^2 =$ constant. Differentiation with 11.59b gives

$$0 = \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} + \mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = 2\left(\mathbf{v} \cdot \frac{d\mathbf{v}}{dt}\right).$$

But this implies that $\mathbf{v}$ and $d\mathbf{v}/dt$ are perpendicular.

Important Remark

Geometrically, this result says that if a curve lies on a sphere with center at the origin (i.e. all position vectors on the curve have the same length), then the tangent vector $\vec{r}\,'(t)$ is always perpendicular to the position vector $\vec{r}(t)$.

Displacement, Velocity and Acceleration (S.11.13)

Problem 1 [11.13, Prob. 17]:

A particle travels around the circle $x^2 + y^2 = 4$ counterclockwise at a constant speed, making 2 revolutions each second. If x and y are measured in meters, what is the velocity of the particle when it is at the point $(1, -\sqrt{3})$?

Solution:

The parametric equation of a circle of radius r and angular parameter θ is given below.

$$x = r\cos(\theta)$$
$$y = r\sin(\theta)$$

The radius of the circle is 2 and the particle makes 2 revolutions per second, so we substitute $\theta = 4\pi t$, where t is a time parameter in seconds.

$$x = 2\cos(4\pi t)$$
$$y = 2\sin(4\pi t)$$

The first t value when the particle is at the point $(1, -\sqrt{3})$ can be found by finding values of t satisfying the system below.

$$1 = 2\cos(4\pi t)$$
$$-\sqrt{3} = 2\sin(4\pi t)$$

The result is $t = \frac{5}{12}$. The velocity of the particle is found by rewriting the parametrization as a vector function and differentiating it with respect to time.

$$\vec{r}(t) = 2\cos(4\pi t)\hat{\boldsymbol{\imath}} + 2\sin(4\pi t)\hat{\boldsymbol{\jmath}}$$

$$\vec{v}(t) = -8\pi\sin(4\pi t)\boldsymbol{\hat{\imath}} + 8\pi\cos(4\pi t)\boldsymbol{\hat{\jmath}}$$

Substituting the t value gives the final velocity at the point $(1, -\sqrt{3})$ in m/s.

$$\vec{v}\left(\frac{5}{12}\right) = -8\pi\sin\left(4\pi\left(\frac{5}{12}\right)\right)\boldsymbol{\hat{\imath}} + 8\pi\cos\left(4\pi\left(\frac{5}{12}\right)\right)\boldsymbol{\hat{\jmath}}$$
$$\vec{v}\left(\frac{5}{12}\right) = 4\pi(\sqrt{3}\boldsymbol{\hat{\imath}} + \boldsymbol{\hat{\jmath}})$$

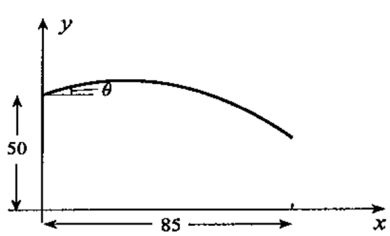
If we choose $t \geq 0$ and $t = 0$ when the particle is at the point $(2, 0)$, its position can be described by $x = 2\cos(4\pi t)$, $y = 2\sin(4\pi t)$. It is at the point $(1, -\sqrt{3})$ when $1 = 2\cos(4\pi t)$, and $-\sqrt{3} = 2\sin(4\pi t)$. This happens for the first time at $t = 5/12$ s. The velocity of the particle at this time is

$$\mathbf{v}(5/12) = -8\pi\sin\left(\frac{5\pi}{3}\right)\hat{\mathbf{i}} + 8\pi\cos\left(\frac{5\pi}{3}\right)\hat{\mathbf{j}} = 4\pi(\sqrt{3}\hat{\mathbf{i}} + \hat{\mathbf{j}}) \text{ m/s.}$$

Problem 2 [11.13, Prob. 33]:

A boy stands on a cliff 50 m high that overlooks a river 85 m wide (see figure page 792 – Textbook Prob. 33). If he can throw a stone at 25 m/s, can he throw it across the river?

Solution:



See the figure above for a sketch of the problem.

The stone can be thrown from any angle θ, and we seek to find if there are any angles where the rock's horizontal displacement exceeds 85m before it falls more than 50m.

The rock is in freefall and experiences an acceleration of $\vec{a} = -g\boldsymbol{\hat{\jmath}}$. Integrating, the velocity of the rock is $\vec{v} = -gt\boldsymbol{\hat{\jmath}} + C$. We use the initial speed of 25m/s to determine C. At $t = 0$, $\vec{v}(0) = C$.

$$\vec{v}(0) = 25(\cos\theta\,\boldsymbol{\hat{\imath}} + \sin\theta\,\boldsymbol{\hat{\jmath}})$$
$$C = 25(\cos\theta\,\boldsymbol{\hat{\imath}} + \sin\theta\,\boldsymbol{\hat{\jmath}})$$

Integrating the velocity gives the position vector function.

$$\vec{v} = -gt\hat{\boldsymbol{j}} + 25(\cos\theta\,\hat{\boldsymbol{\imath}} + \sin\theta\,\hat{\boldsymbol{j}})$$
$$\vec{r} = -\frac{1}{2}gt^2\hat{\boldsymbol{j}} + 25t(\cos\theta\,\hat{\boldsymbol{\imath}} + \sin\theta\,\hat{\boldsymbol{j}}) + D$$

Using the initial condition of $\vec{r}(0) = 50\hat{\boldsymbol{j}}$, $D = 50\hat{\boldsymbol{j}}$. Simplify.

$$\vec{r} = -\frac{1}{2}gt^2\hat{\boldsymbol{j}} + 25t(\cos\theta\,\hat{\boldsymbol{\imath}} + \sin\theta\,\hat{\boldsymbol{j}}) + 50\hat{\boldsymbol{j}}$$

$$\vec{r} = (25t\cos\theta)\hat{\boldsymbol{\imath}} + \left(-\frac{1}{2}gt^2 + 25t\sin\theta + 50\right)\hat{\boldsymbol{j}}$$

When the stone hits the water, $y = 0$. Solving for t gives the time when the stone hits the water.

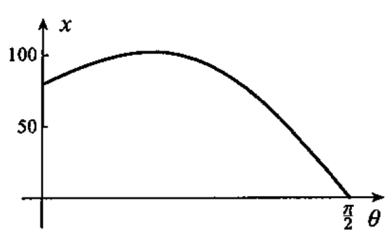
$$-\frac{1}{2}gt^2 + 25t\sin\theta + 50 = 0$$

$$t = \frac{25\sin\theta \pm \sqrt{625\ \sin(\theta)^2 + 100g}}{g}$$

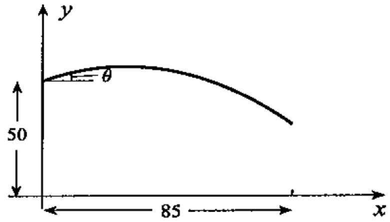
Using the positive solution, substituting this t into the x-coordinate of the position vector gives the x-coordinate when the stone hits the water in terms of the angle it is thrown at, θ.

$$x = 25t\cos\theta$$
$$x = 25\cos\theta\left(\frac{25\sin\theta + \sqrt{625\ \sin(\theta)^2 + 100g}}{g}\right)$$

A plot of this function shows that some throwing angles result in an x displacement greater than 85m. The stone can be thrown across the river.



The acceleration of the stone is $\mathbf{a} = -g\hat{\mathbf{j}}$, so that $\mathbf{v} = -gt\hat{\mathbf{j}} + \mathbf{C}$. If we take as $t = 0$ the time when the stone is thrown, then when it is thrown at angle θ,



$$\mathbf{v}(0) = 25(\cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}}),$$

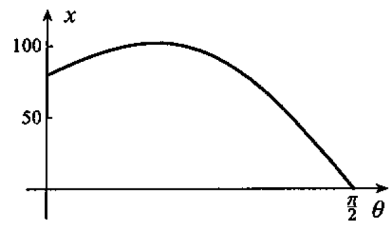
from which

$$25(\cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}}) = \mathbf{C}.$$

Integration of $\mathbf{v} = -gt\hat{\mathbf{j}} + \mathbf{C}$ gives $\mathbf{r} = -gt^2\hat{\mathbf{j}}/2 + \mathbf{C}t + \mathbf{D}$.
Since $\mathbf{r}(0) = 50\hat{\mathbf{j}}$, it follows that $\mathbf{D} = 50\hat{\mathbf{j}}$, and

$$\mathbf{r} = -\frac{1}{2}gt^2\hat{\mathbf{j}} + 25t(\cos\theta\,\hat{\mathbf{i}} + \sin\theta\,\hat{\mathbf{j}}) + 50\hat{\mathbf{j}} = (25t\cos\theta)\hat{\mathbf{i}} + \left(-\frac{1}{2}gt^2 + 25t\sin\theta + 50\right)\hat{\mathbf{j}}.$$

The stone hits water level when $y = 0$ in which case



$$0 = -\frac{1}{2}gt^2 + 25t\sin\theta + 50.$$

from which

$$t = \frac{-25\sin\theta \pm \sqrt{625\sin^2\theta + 100g}}{-g}.$$

Using the positive solution, the x-coordinate of the stone at water level is

$$x = 25\cos\theta\left(\frac{25\sin\theta + \sqrt{625\sin^2\theta + 100g}}{g}\right).$$

A plot of this function shows that there are indeed angles for which $x \geq 85$.

Length of Curves (S.11.11)

Problem 1 [S. 11.11, Prob. 11]:

Find the length of the following curve. Draw the curve.

$$x = 2\cos t, \qquad y = 2\sin t, \qquad z = 3t, \qquad 0 \le t \le 2\pi$$

Solution:

The curve is a helix lying in a circular cylinder of radius 2 and traveling upwards. The position vector can be written as,

$$\vec{r}(t) = 2\cos t\,\hat{\imath} + 2\sin t\,\hat{\jmath} + 3t\hat{k}$$

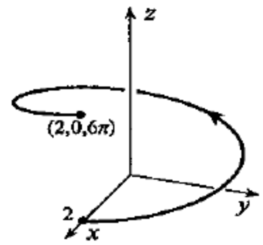
The speed can be obtained from,

$$|\vec{v}| = \left|\frac{d\vec{r}}{dt}\right| = \left|-2\sin t\,\hat{\imath} + 2\cos t\,\hat{\jmath} + 3\hat{k}\right|$$

The length of the curve is therefore given by,

$$L = \int_{t=\alpha}^{t=\beta} |\vec{v}(t)| dt$$

$$L = \int_0^{2\pi} \sqrt{(-2\sin t)^2 + (2\cos t)^2 + 9}\, dt$$
$$= \sqrt{13}\left\{t\right\}_0^{2\pi} = 2\sqrt{13}\pi.$$



Problem 2 [S. 11.11, Prob. 13]:

Find the length of the following curve. Draw the curve.

$$x = t^3, \qquad y = t^2, \qquad z = t^3, \qquad 0 \le t \le 1$$

Solution:

The curve can be drawn by noticing that it is the intersection of the plane $x = z$ and the cylinder $y = x^{2/3}$.

The position vector can be written as,

$$\vec{r}(t) = t^3\hat{\imath} + t^2\hat{\jmath} + t^3\hat{k}$$

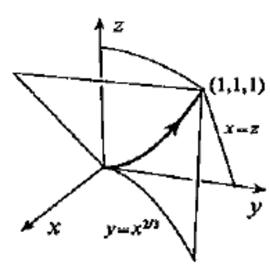
The speed can be obtained from,

$$|\vec{v}| = \left|\frac{d\vec{r}}{dt}\right| = \left|3t^2\hat{\imath} + 2t\hat{\jmath} + 3t^2\hat{k}\right|$$

The length of the curve is therefore given by,

$$L = \int_{t=\alpha}^{t=\beta} |\vec{v}(t)| dt$$

$$L = \int_0^1 \sqrt{(3t^2)^2 + (2t)^2 + (3t^2)^2}\, dt$$
$$= \int_0^1 t\sqrt{4 + 18t^2}\, dt.$$
$$= \left\{\frac{(4 + 18t^2)^{3/2}}{54}\right\}_0^1 = \frac{11\sqrt{22} - 4}{27}.$$



Tangent Vectors (S.11.11)

Problem 1 [S. 11.11, Prob. 5]:

Express the following curve in vector form and find the unit tangent vector $\widehat{\mathbf{T}}$ at each point in the curve:

$$x + y + z = 4, \quad x^2 + y^2 = 4, \ \ y \geq 0$$

going from $(2,0,2,)$ to $(-2,0,6)$

Suggestion: Sketch the curve for practice

Solution:

Using the parametrization,

$$x = 2\cos t$$

$$y = 2\sin t$$

Then,

$$z = 4 - 2\cos t - 2\sin t$$

If the curve is going from $(2,0,2,)$ to $(-2,0,6)$, the range of t is $0 \leq t \leq \pi$. A vector representation of the curve is then,

$$\vec{r}(t) = 2\cos t\,\hat{\imath} + 2\sin t\,\hat{\jmath} + (4 - 2\cos t - 2\sin t)\hat{k}$$

The tangent vector is then,

$\mathbf{T} = \frac{d\mathbf{r}}{dt} = -2\sin t\,\hat{\mathbf{i}} + 2\cos t\,\hat{\mathbf{j}} + (2\sin t - 2\cos t)\hat{\mathbf{k}}$, and a unit tangent vector is

$$\hat{\mathbf{T}} = \frac{-2\sin t\,\hat{\mathbf{i}} + 2\cos t\,\hat{\mathbf{j}} + (2\sin t - 2\cos t)\hat{\mathbf{k}}}{\sqrt{4\sin^2 t + 4\cos^2 t + (2\sin t - 2\cos t)^2}} = \frac{-\sin t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}} + (\sin t - \cos t)\hat{\mathbf{k}}}{\sqrt{2 - \sin 2t}}.$$

Problem 2 [S. 11.11, Prob. 9]:

Find $\widehat{T}$ at the point $(5,2,10)$ of the curve $x = y^2 + 1, \ \ z = x + 5$, directed so that y increases along the curve.

Suggestion: Sketch the curve for practice

Solution:

Since we need the curve directed so that y increases along the curve, we use the parametrization,

$$y = t$$

That way, y increases with the ascending parameter t. Then,

$$x = t^2 + 1$$

$$z = t^2 + 6$$

A vector representation of the curve is then,

$$\vec{r}(t) = (t^2 + 1)\hat{\imath} + t\hat{\jmath} + (t^2 + 6)\hat{k}$$

With $x = t^2 + 1$, $y = t$, $z = t^2 + 6$, a tangent vector is $\dfrac{d\mathbf{r}}{dt} = 2t\hat{\mathbf{i}} + \hat{\mathbf{j}} + 2t\hat{\mathbf{k}}$. A unit tangent vector at $(5, 2, 10)$ is $\hat{\mathbf{T}}(2) = \dfrac{4\hat{\mathbf{i}} + \hat{\mathbf{j}} + 4\hat{\mathbf{k}}}{\sqrt{33}}$.

Curvature and Radius of Curvature (S.11.12)

Problem 1 [11.12, Probs. 15]:

Find the curvature and the radius of curvature of the curve (if it exists). Draw the curve.

$$x = t, \qquad y = t^3, \qquad z = t^2, \quad t \geq 0$$

Solution:

The curve can be drawn by noticing that it is the intersection of the cubic cylinder $y = x^3$ and the parabolic cylinder $z = x^2$.

The position vector can be written as,

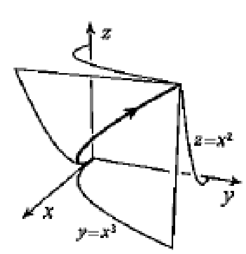
$$\vec{r}(t) = t\hat{\imath} + t^3\hat{\jmath} + t^2\hat{k}$$

Which can be differentiated once to obtain $\vec{r}'(t)$, and twice to obtain $\vec{r}''(t)$. The curvature is therefore given by,

$$\kappa(t) = \frac{|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}|}{|\dot{\mathbf{r}}|^3} = \frac{|(1, 3t^2, 2t) \times (0, 6t, 2)|}{|(1, 3t^2, 2t)|^3}$$

$$= \frac{1}{(1 + 4t^2 + 9t^4)^{3/2}} \left\| \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 3t^2 & 2t \\ 0 & 6t & 2 \end{vmatrix} \right\|$$

$$= \frac{|(-6t^2, -2, 6t)|}{(1 + 4t^2 + 9t^4)^{3/2}} = \frac{2\sqrt{1 + 9t^2 + 9t^4}}{(1 + 4t^2 + 9t^4)^{3/2}}$$



The radius of curvature will therefore be given by,

$$\rho(t) = \frac{1}{\kappa} = \frac{(1+4t^2+9t^4)^{3/2}}{2\sqrt{1+9t^2+9t^4}}.$$

Problem 2 [11.12, Probs. 17]:

Find the curvature and the radius of curvature of the curve (if it exists). Draw the curve.

$$x = t+1, \qquad y = t^2 - 1, \qquad z = t+1, \quad -\infty < t < \infty$$

Solution:

If $x = t+1$, then $t = x-1$, and

$$y = (x-1)^2 - 1$$

$$y = x^2 - 2x + 1 - 1 = x(x-2)$$

The curve can be drawn by noticing that it is the intersection of the parabolic cylinder $y = x(x-2)$ and the plane $z = x$.

The position vector can be written as,

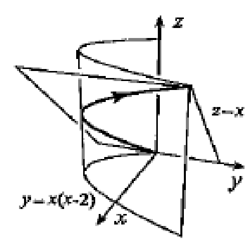
$$\vec{r}(t) = (t+1)\hat{\imath} + (t^2-1)\hat{\jmath} + (t+1)\hat{k}$$

Which can be differentiated once to obtain $\vec{r}'(t)$, and twice to obtain $\vec{r}''(t)$. The curvature is therefore given by,

$$\kappa(t) = \frac{|\dot{\mathbf{r}} \times \ddot{\mathbf{r}}|}{|\dot{\mathbf{r}}|^3} = \frac{|(1,2t,1)\times(0,2,0)|}{|(1,2t,1)|^3} = \frac{1}{(2+4t^2)^{3/2}} \left\| \begin{matrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2t & 1 \\ 0 & 2 & 0 \end{matrix} \right\|$$

$$= \frac{|(-2,0,2)|}{(2+4t^2)^{3/2}} = \frac{1}{(1+2t^2)^{3/2}}$$

and $\rho(t) = \dfrac{1}{\kappa} = (1+2t^2)^{3/2}$.



Normal Vectors, Curvature and Radius of Curvature (S.11.12)

Problem 1 [11.12, Prob. 9] :

Find $\widehat{N}$ and $\widehat{B}$ at the point (5,2,10) on the curve: $x = y^2 + 1$, $z = x + 5$, directed so that y increases along the curve.

Solution:

Since we need the curve directed so that y increases along the curve, we use the parametrization, we can use the same aproach for **Problem 2 [S. 11.11, Prob. 9]** above,

$$y = t$$

That way, y increases with the ascending parameter t. Then,

$$x = t^2 + 1$$

$$z = t^2 + 6$$

A vector representation of the curve is then,

$$\vec{r}(t) = (t^2 + 1)\hat{\imath} + t\hat{\jmath} + (t^2 + 6)\hat{k}$$

With parametric equations $x = t^2 + 1$, $y = t$, $z = t^2 + 6$, a unit tangent vector to the curve is $\hat{\mathbf{T}} = \dfrac{(2t, 1, 2t)}{\sqrt{8t^2 + 1}}$. A vector in the direction of $\hat{\mathbf{N}}$ is

$$\mathbf{N} = \frac{d\hat{\mathbf{T}}}{dt} = \frac{-8t}{(1 + 8t^2)^{3/2}}(2t, 1, 2t) + \frac{(2, 0, 2)}{\sqrt{1 + 8t^2}}.$$

At $(5, 2, 10)$, $t = 2$, in which case

$$\mathbf{N}(2) = \frac{-16}{33\sqrt{33}}(4, 1, 4) + \frac{(2, 0, 2)}{\sqrt{33}} = \frac{2(1, -8, 1)}{33\sqrt{33}}.$$

Hence, the principal normal at $(5, 2, 10)$ is $\hat{\mathbf{N}} = \dfrac{(1, -8, 1)}{\sqrt{66}}$. Since a tangent vector at the point is $\mathbf{T}(2) = (4, 1, 4)$, the direction of the binormal at the point is

$$\mathbf{B}(2) = (4, 1, 4) \times (1, -8, 1) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 4 & 1 & 4 \\ 1 & -8 & 1 \end{vmatrix} = (33, 0, -33).$$

Thus, $\hat{\mathbf{B}}(2) = (1, 0, -1)/\sqrt{2}$.

Problem 2: [11.12, Prob. 23]

Let C be the curve $x = t, y = t^2$ in the xy-plane.

a. At each point on C calculate the unit tangent $\hat{T}$ vector and the principal normal $\hat{N}$. What is $\hat{B}$?
b. $\vec{F} = t^2\hat{\imath} + t^4\hat{\jmath}$ is a vector that is defined at each point P on C. Denote by F_T and F_N the components of $\vec{F}$ in the directions $\hat{T}$ and $\hat{N}$ respectively. Find F_T and F_N as functions of t.
c. Express F in terms of $\hat{T}$ and $\hat{N}$.

Solution:

(a) From the given parametric equations, we can find

$$\hat{T} = \frac{(1, 2t)}{\sqrt{1+4t^2}}$$

The unit normal vector can be calculated as

$$\hat{\mathbf{N}} = \frac{\frac{d\hat{\mathbf{T}}}{dt}}{\left|\frac{d\hat{\mathbf{T}}}{dt}\right|}$$

Where,

$$\frac{d\hat{\mathbf{T}}}{dt} = \frac{(-4t, 2)}{(1+4t^2)^{\frac{3}{2}}}$$

$$\left|\frac{d\hat{\mathbf{T}}}{dt}\right| = \sqrt{\frac{16t^2+4}{(1+4t^2)^3}} = 2\sqrt{\frac{1+4t^2}{(1+4t^2)^3}} = 2\frac{\sqrt{1+4t^2}}{(1+4t^2)^{\frac{3}{2}}}$$

$$\hat{\mathbf{N}} = \frac{\frac{(-4t,2)}{(1+4t^2)^{\frac{3}{2}}}}{2\frac{\sqrt{1+4t^2}}{(1+4t^2)^{\frac{3}{2}}}} = \frac{(-4t,2)}{2\sqrt{1+4t^2}} = \frac{(-2t,1)}{\sqrt{1+4t^2}}$$

The direction of the binormal is $\mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2t & 0 \\ -2t & 1 & 0 \end{vmatrix} = (0, 0, 1+4t^2)$. Thus, $\hat{\mathbf{B}} = (0,0,1)$.

(b) $F_T = \mathbf{F} \cdot \hat{\mathbf{T}} = (t^2, t^4) \cdot \frac{(1,2t)}{\sqrt{1+4t^2}} = \frac{t^2+2t^5}{\sqrt{1+4t^2}}$; $F_N = \mathbf{F} \cdot \hat{\mathbf{N}} = (t^2, t^4) \cdot \frac{(-2t,1)}{\sqrt{1+4t^2}} = \frac{t^4-2t^3}{\sqrt{1+4t^2}}$

(c) $\mathbf{F} = F_T\hat{\mathbf{T}} + F_N\hat{\mathbf{N}} = \frac{t^2+2t^5}{\sqrt{1+4t^2}}\hat{\mathbf{T}} + \frac{t^4-2t^3}{\sqrt{1+4t^2}}\hat{\mathbf{N}}$

Extra Practice Problems[1]

The answers to these problems, along with all even-numbered exercises, are provided at the back of the textbook. Detailed solutions can be found in Trim's Student Solution Manual. If you don't own a copy, a reserved edition is available at the Davis Centre.

1. S. 11.10, Probs. 6, 14
2. S. 11.9, Probs. 10, 18, 24
3. S. 11.11, Probs. 4, 10, 12, 14
4. S. 11.12, Probs. 10, 18, 24
5. S. 11.13, Probs. 12, 14, 24

[1] You are strongly encouraged to try and solve the problems assigned in this homework on your own, before looking at the solutions. These problems are only the minimum necessary to master the material. You are also encouraged to do additional problems from the text for practice (some suggested **extra practice problems** have been listed here).